

Vipul Kumar

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EDUCATION

Vellore Institute of Technology | Vellore

2023 – 2027

B.Tech in Computer Science Engineering (Core)

GPA: 8.43/10.00

PROFESSIONAL SUMMARY: Computer Science Engineering student with experience in Full Stack Development, Generative AI, Machine Learning, and Computer Vision, focused on building scalable and AI-powered solutions.

INTERNSHIPS

[Project Head | Computer Vision and ML Intern](#) | [MarkMyAd](#)

July 2026 – Present

- Developed real-time video analytics pipelines using YOLO, DeepSORT, and OpenCV for object detection and multi-object tracking.
- Optimized multi-object tracking pipelines and integrated deep learning models for efficient real-time inference.
- Built computer vision solutions for real-time video analytics, enabling automated object detection and tracking.
- Led computer vision and ML development, driving model integration, testing, and deployment of production systems.

[Software Engineer Intern \(Generative AI\)](#) | [KreditBee](#) | [Certificate](#)

May 2026 – July 2026

- Engineered Gemini-powered AI voice agent for automated customer conversations with context-aware LLM responses.
- Developed an AI user simulator and LLM evaluator to automate conversation testing in diverse customer scenarios.
- Benchmarked AI responses against real interactions using embeddings, UMAP, and conversational similarity analysis.

[Full Stack Development Intern](#) | [Valsco \(Jurident\)](#) | [Certificate](#)

Feb 2026 – May 2026

- Built the full-stack application, developing its frontend with React and backend using Node.js, FastAPI, and Firebase.
- Developed and optimized RESTful APIs and database queries, ensuring low latency, scalability, and efficient data flow.

[Software Development Intern](#) | [Bluestock Fintech](#) | [Certificate](#)

June 2025 – July 2025

- Made a full-stack IPO web app using React.js, Node.js, and PostgreSQL with responsive UI and REST APIs.
- Implemented PostgreSQL schema optimization and Git workflows across application development and deployment.

HACKATHONS

- [SIH \(Internal Finalist\) | 2024](#) : Integrated a women's safety and threat detection system using YOLO, computer vision, and ML to identify assault, harassment, fighting, and high-risk interactions from video streams, achieving ~97% detection accuracy and triggering real-time alerts with timestamps, evidence snapshots, and incident information.
- [National Space Hackathon – IIT Delhi](#) : Architected an AI-driven space debris collision prediction system that processes orbital data to track debris trajectories, assess satellite collision probability, detect high-risk encounters, and provide data-driven recommendations for collision avoidance and satellite safety.
- [Code for Bharat – Season 2](#) : AI-powered missing person detection system using computer vision and CCTV video streams to identify registered individuals in real time, achieving ~96% detection accuracy and automatically generating alerts with captured evidence and person details when a potential match was detected.

PROJECTS

[AI Career Copilot](#) | Tech Stack: Next.js, FastAPI, Gemini, PostgreSQL, Redis, Docker Feb 2026 - Present

- Engineered a startup-grade AI career SaaS platform integrating AI resume building and scoring, GitHub analysis, coding practice, mock interviews, personalized job matching, AI mentoring, and subject-wise learning resources.

[Hostel Management System](#) | Tech Stack: Kotlin, Jetpack Compose, Firebase Authentication Feb 2026

- Implemented a native Android hostel management app with Google authentication, hostel profiles, and maintenance ticket tracking for cleaning, electrical, plumbing, and AC services.

[ML-Based Fruit Quality Evaluation](#) | Tech Stack: Python, ML, Computer Vision

Dec 2025

- Developed a computer vision-based system to classify fruit quality, improving inspection accuracy using ML models.

[Real-Time Fire and Smoke Detection](#) | Tech Stack: Python, OpenCV, YOLO

Aug 2025

- Created a real-time fire and smoke detection system using YOLOv8 and OpenCV with severity scoring.

[E-Commerce](#) and [Food Ordering Platform](#) | Tech Stack: MERN stack

Mar 2025

- Built a full-stack shopping platform supporting browsing, live tracking, orders, payments, and authentication.

TECHNICAL SKILLS

- **Frontend and Backend:** React, Next.js, Tailwind CSS, HTML, CSS, JavaScript, Node.js, Express.js, FastAPI
- **Languages and Tools:** Python, Java, C++, C, MongoDB, PostgreSQL, MySQL, APIs, GitHub, Vercel, Render
- **Machine Learning & Data Science:** Computer Vision, Scikit-Learn, NumPy, TensorFlow, OpenCV, YOLO, Matplotlib, Jupyter Notebook, Pandas, Embeddings, UMAP
- **Generative AI:** Gemini API, LLMs, AI Agents, Prompt Engineering, LangChain, RAG, Voice AI

CERTIFICATES

- [Machine Learning Specialization](#) – DeepLearning.AI & Stanford Online (Andrew Ng)
- [Advanced Generative AI](#) – IBM Career Education Program (IBMCE)